Mars

The Red Planet - fourth planet from the Sun

Overview

Mars is the fourth planet from the Sun and the second-smallest planet in the Solar System, after Mercury. It is named after the Roman god of war. Its reddish colour comes from iron oxide (rust) in the dust and rocks that cover its surface.

Mars is a terrestrial (rocky) planet. Of all the planets, it is the one most studied as a possible place for past microbial life and future human exploration, because water ice exists at its poles and below its surface.

Key facts

- **Average distance from the Sun:** 1.52 AU (about 228 million km)
- **Diameter:** 6,779 km (about 53% of Earth)
- **Length of day:** 24 hours 37 minutes (one Martian day is called a 'sol')
- **Length of year:** 687 Earth days
- **Surface gravity:** 3.71 m/s squared (about 38% of Earth)
- **Average temperature:** about -63 C
- **Number of moons:** 2 (Phobos and Deimos)
- **Axial tilt:** 25.2 degrees

Atmosphere and climate

The Martian atmosphere is very thin: surface pressure is less than 1% of Earth's. It consists of about 95% carbon dioxide, with small amounts of nitrogen and argon. Because its axial tilt is similar to Earth's, Mars has seasons, but each season lasts roughly twice as long because the Martian year is longer.

Mars is known for dust storms. Local storms are common, and every few years a storm can grow to cover almost the entire planet. A global dust storm in 2018 blocked sunlight and ended the mission of the solar-powered Opportunity rover.

Surface features

Mars hosts **Olympus Mons**, the tallest volcano known in the Solar System. It rises about 22 km above the surrounding plains, roughly two and a half times the height of Mount Everest.

Valles Marineris is an enormous canyon system about 4,000 km long and up to 7 km deep. On Earth it would stretch across the entire continental United States. Both polar regions have ice caps made of water ice covered by seasonal layers of frozen carbon dioxide.

Moons

Mars has two small, irregularly shaped moons, **Phobos** and **Deimos**, discovered by the American astronomer Asaph Hall in 1877. They may be captured asteroids. Phobos orbits so close to Mars that it is slowly spiralling inward and is expected to break apart or crash into the planet in tens of millions of years.

Interior and magnetic field

Mars has a crust, a mantle and a core made mostly of iron, nickel and sulfur. Seismic data from NASA's **InSight** lander (2018-2022), which recorded more than 1,300 'marsquakes', showed that the core is at least partly liquid.

Unlike Earth, Mars no longer has a global magnetic field. It is thought to have lost it around 4 billion years ago. Without this protective shield, the solar wind gradually stripped away much of the planet's atmosphere, which is one reason Mars became cold and dry.

Water on Mars

Today liquid water cannot remain stable on the surface for long because the air pressure is so low. However, there is strong evidence that Mars was once much wetter: orbiters and rovers have found dried river valleys, ancient lake beds, river deltas and minerals that only form in water. Jezero Crater, where Perseverance landed, was once a lake fed by a river delta.

Large amounts of water ice still exist in the polar caps and frozen in the ground at middle latitudes.

Observing Mars from Earth

Mars is easily visible to the naked eye as a reddish 'star'. About every 26 months Earth passes between Mars and the Sun - an event called **opposition** - when Mars appears brightest and is closest to us. In August 2003 Mars came within about 55.8 million km of Earth, its closest approach in nearly 60,000 years.

The same 26-month cycle sets the 'launch windows' for missions to Mars. With current rockets, the trip takes about seven months.

Exploration history

- **Mariner 4** (NASA, 1965) - first successful flyby; returned the first close-up photos of another planet.
- **Viking 1 and 2** (NASA, 1976) - first successful long-term landers; searched for signs of life.
- **Spirit and Opportunity** (NASA, landed 2004) - twin rovers; Opportunity operated until 2018.
- **Curiosity** (NASA, landed August 2012) - car-sized rover exploring Gale Crater.
- **Perseverance** (NASA, landed 18 February 2021) - rover in Jezero Crater collecting rock samples for future return to Earth.
- **Ingenuity** - small helicopter carried by Perseverance; on 19 April 2021 it made the first powered, controlled flight on another planet.
- **Zhurong** (China, landed May 2021) - China's first Mars rover.

Did you know?

- Mars has the largest dust storms in the Solar System.
- Sunsets on Mars appear blue because of the way fine dust scatters light.
- A person weighing 70 kg on Earth would feel like they weigh about 27 kg on Mars.